Discover, Observe, Explore!

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This worksheet is suitable for classes 6 to 8.

Learning to classify through observation

(Class 6)*

Skills addressed: Observation, Analysis, Reasoning.

Scientists classify objects in Nature into different groups called taxa. How do they go about doing this? In order to classify animals and plants, scientists always begin by observing them, noting their characteristics or behaviour and then picking them up by hand to know more about them. Would you like to be one such scientist? Do the following activity and learn about classification through observation.

1. Placed in front of you is a bowl with different types of peas/lentils (dals). You are not permitted to touch them or pick them up. You may use a spoon **once** to move them around, so that you don't miss the peas/lentils that may be hidden below.
2. Your first task is to spend time just observing the peas/lentils.
3. Are they all the same or are there different kinds of peas/lentils? [Look for qualities that are common or different between them.]

   Answer: _____
4. Based on your observations, sort them into different sets. [You are allowed to separate them into as many sets as you want, but you will have to give reasons for doing so. Remember each set into which you have separated them must have similar characteristics.] How many sets of peas/lentils are there in the bowl?

   Answer: _____
5. Note down the characteristics (qualities) observed and write them down on this paper. You can draw up a table:

Set No.	Qualities (characteristics)				

   or simply put them down as:

   Set 1: _____

   Set 2: _____

   Set 3: _____

   Set 4: _____

   Set 5: _____

   Depending on how many sets you have sorted the peas/lentils into, you may add more sets or columns and rows. Against each set, write down your observations.
   [For example, if I were observing birds, I would write Set 1: the size of a sparrow; brown in colour; round head; sharp beak]

6. After sorting the peas/lentils [i.e., having classified them] into sets, give a name to each set based on the qualities you observed.

   Set 1: _____

*Teachers Resource Sheet for this activity is given on page 34.

Set 2: _____

Set 3: _____

Set 4: _____

Set 5: _____

Your teacher has put up a paper on the notice board. Look at it. It has images of the peas/lentils in the bowl.

7. Did your observation help you sort them into the same groups as shown in the images?

   Answer: _____

8. Why did your classification match/not match?

   Answer: _____

9. If your classification did not match, what should you do to correct your mistakes? Given below are choices for your next step. Tick the correct choice/choices.
   i) Pick them up in your hand and examine them. ☐
   ii) Look at the pods (fruits that contain these seeds). ☐
   iii) Search and find the plants that produced these peas/lentils. ☐
   iv) Go and look for them in the kitchen. ☐
   v) Search for information in the library. ☐
   vi) Search the internet for information. ☐
   vii) Ask your teacher for help. ☐

10. You may have seen these peas/lentils in your kitchen at home. Do you know their names?

Novel ways to recycle materials

(Class 7)

Time: 1 hour.
Skills addressed:
1. **Application:** Using information in new situations for solving problems.
2. **Creation:** Producing new or original work by combining elements in novel ways.

We often hear about the need to reuse and recycle materials in order to reduce pollution. Give wings to your imagination to solve the problem of pollution from non-biodegradable materials, which are generally thrown out as waste.

Given below is a list of a few non-biodegradable materials. Against each item, write at least one way in which it can be meaningfully reused. You may give (if you can find or know) more than one way of reusing the material.

1. Small plastic packets

2. Ice cream spoons

3. Crepe bandage container

4. Plastic boxes of different sizes

5. Bathing soap wrappers (paper and plastic)

6. Used pen refill

7. Empty shampoo bottles

8. Bubble wrap

9. Grey colour plastic bags

10. Worn out toothbrush

Balanced diet

(Class 7)*

Skills: Connecting facts to build knowledge, Self-study.

You have to refer to the chart on the notice board, which has the calorie value of different foods to answer these questions.

1. Your mother is recovering from a bout of fever. The doctor has advised her to have a diet that will give her an extra 800-1000 calories per day and foods rich in proteins and vitamins for one week. He has also advised her not to have too much of fatty foods. Draw up a balanced diet for your mother for one week.

2. Your sister is practicing hard to be selected as a long-distance runner for the Asian Games. Her coach has recommended that she get 60% to 70% of her calories from carbohydrates, with lean protein and healthy fats each accounting for 15% to 20% of their remaining calories. He also advised that calcium and vitamin D are essential.

   i) Why did he highlight that calcium and Vitamin D are essential? What should her diet contain so she gets these nutrients in adequate quantities?

   ii) Suggest a suitable diet for her everyday consumption.

3. Your 75 year-old grandfather is a healthy and active man, who goes for walks every morning and practices yoga regularly. What is his food regimen, which keeps him healthy and energetic? List the constituents of his diet.

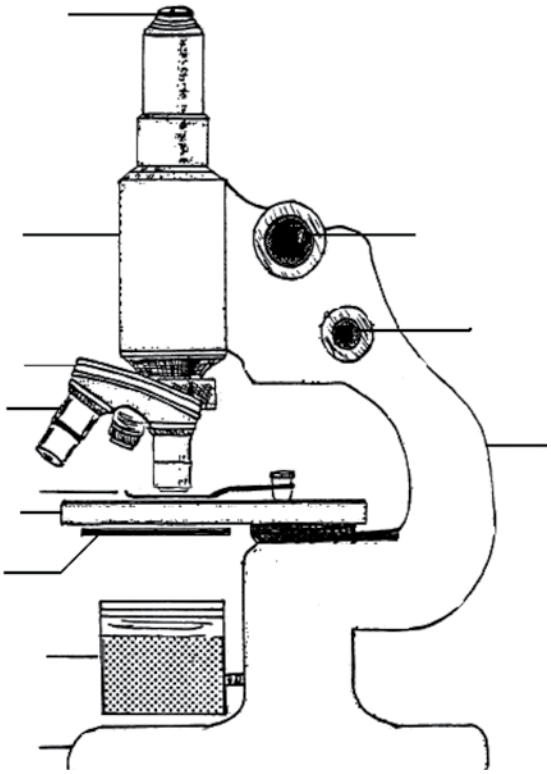
*The food-calorie chart on page 35 should be displayed on the notice board so that students can refer and answer the questions above.

Parts of a microscope

(Class 8)*

Skills: Self-study, Reasoning, Handling instruments.
Time: 40 minutes

A microscope allows you to observe things that cannot be seen clearly by the naked eye. As its name suggests, a microscope is an instrument used to see the 'micro' or the world of small beings.



I. Mark on the diagram the parts given below.

1. Ocular lens / eyepiece:
2. Body tube:
3. Arm:
4. Coarse adjustment knob:
5. Fine adjustment knob:
6. Revolving nosepiece:
7. Objective lens:
8. Stage:
9. Stage clips:
10. Diaphragm:
11. Light source:

Write down their functions against each part.

II. A teacher gives a glass slide that has a section of a leaf on it. She asks her class to observe it under the microscope and tell her what they are able to see. Some students faced problems while using the microscope. Take a look at the problems they ran into and give reason(s) why.

a) Ankit could not see the glass slide when he looked into the microscope. Why?

Answer: _____

b) Every time Akanksha tilted the microscope to see through the eyepiece, the glass slide fell down. How will you solve this problem for her?

Answer: _____

c) Archana put the slide on the microscope stage, brought down the body tube and brought into focus what was on the slide. But everything is dark and so she could not see the parts of the leaf. She soon solved the problem and got the light to fall on the slide. What did she do?

Answer: _____

d) Abhishek could see small openings on the leaf, but he could not see their shape or size? What should he do to see more details?

Answer: _____

* Teachers Resource Sheet for this activity is given on page 37.

Insects and breathing
(Class 8)

Skills addressed: Thinking, Reasoning, Making connections, and Inferring to conclude.
Time: 40 minutes

Your teacher has placed in front of you a small glass bottle (15 ml by volume) with 2-3 bow flies (these are flies you will see on animal dung and human feces. They are blue/green, shiny in appearance) and a few ants. She has covered the mouth of the bottle with a cork through which a blue litmus paper has been inserted such that a part of the paper is inside the glass jar and a part of it outside. Answer the following questions based on the setup by your teacher.

1. What is blue litmus paper generally used for?

   _____

2. How is a blue litmus paper made?

   _____

3. Observe the insects inside the bottle. They may be moving slowly or staying still. You will see that a small part of the blue litmus paper is beginning to turn pink. Why?

   _____

4. After five minutes, one of you may shake the bottle gently. Do not shake vigorously. Put it down and observe. The flies sensing danger begin to move about inside the bottle. The ants too get active. The blue litmus paper now begins to turn red more rapidly. Why?

   _____

5. If you have difficulty answering the question above then here is a helpful hint. Close your mouth and breathe through your nose. When you exhale, keep a blue litmus paper near the nose, let the air you exhale fall on the blue litmus paper. Do you see it changing to red? Why? Now, answer the questions connected with the insects above.

   _____

6. What is common between you and the insects that the blue litmus paper turns red in both cases?

   _____

Fingerprints
(Class 8)

Skills: Observation, Analysis, and Inference.

Fingerprints are considered the most authentic evidence of a person's identity, especially when disputes arise. If you want your passport made, in addition to all other documents, your fingerprints are registered. You can see your prints if you press your finger (any one of the five) on an inkpad first and then press it down on a thick white sheet. It's fun to identify the types of fingerprints you possess.

Fingerprints are basically classified into three groups – arches, loops, and whorls. Each of these types are further classified into five smaller groups. Totally, 8 types of fingerprints can be identified; they are given below for you to identify. How will you do it?

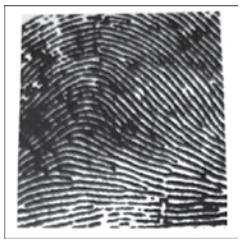
To begin with, you have to look for deltas and count how many deltas are present. A delta is a place where the fingerprint ridges make a small triangle. See the image below – the red marks highlight the delta.

If a fingerprint has no deltas, then it's an **arch**. If it is just one delta, then it's a **loop**. If the loop points towards the thumb, then it's called a radial loop. If it points towards the little finger then it's called the ulnar loop. The most common fingerprint is the ulnar loop. 65% of people have the ulnar loop. Two deltas mean it's a **whorl**.

Given below are the eight different kinds of fingerprints. Also given are the eight names by which each fingerprint is described. Study the print images and match them correctly with the names given.

Fingerprint image

1.

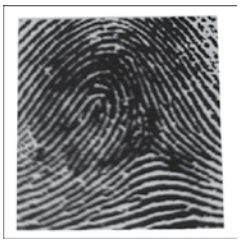


_____

2.

_____

3. [towards thumb]

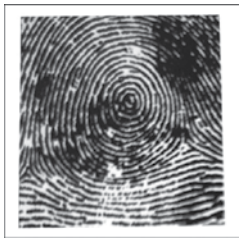


_____

4. [towards little finger]

_____

5.



_____

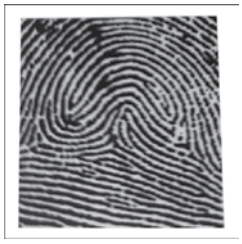
6.

_____

7.

_____

8.



_____

Accidental whorl; Plain arch; Ulnar loop; Double loop whorl; Tented arch loop; Plain whorl; Radial loop, Central pocket whorl

Have fun!

Learning to classify through observation

(Class 6)

Teacher Information and Resource Sheet

I. Time: 40 minutes.

II. Type of activity: Pen and paper activity.

III. Skills addressed: Observation, Analysis, Reasoning, and Writing.

IV. Materials: Different types of lentils/peas listed below. Each lentil sample is only to be numbered and their names not to be displayed on the notice board.

1. Chickpeas or Bengal gram whole
2. Split chickpeas or split Bengal gram
3. Roasted chickpeas or roasted Bengal gram
4. Split and husked green gram
5. Pigeon pea

V. The list of local names given under **Reference** may be printed out and displayed on the class notice board for students to refer to after the activity is over.

VI. Students are to be grouped at the beginning of the activity with each group not having more than six students.

VII. Each group is to be provided with a bowl that contains a mixture of the five lentils/peas listed above and then given the worksheet to be filled. Although working in groups, the worksheet is to be given to each student to be filled individually. They have to choose a group leader who will read out the findings of the group at the end of the activity.

VIII. Following the presentation from the groups, share the paper with the images of the five peas/lentils presented to them as a mixture. Did their classification match the five groups of peas/lentils?

IX. Have a short discussion about what the next step should be to classify the peas/lentils correctly.

X. Explain classification and its importance briefly.

Images of lentils/peas for students to refer to at the conclusion of the activity.

1.
2.
3.
4.
5.

Reference: Given below are local Indian names for the lentils/peas from the activity.

1. English: Bengal gram whole/chickpea
   Hindi: Chana, Marathi: Harbara, Bengali: Chola, Gujarati: Chana, Punjabi: Chole, Kashmiri: Chanu, Oriya: Buta, Tamil: Konda kadalai, Malayalam: Kadala, Telugu: Senagalu, Kannada: Kadale
2. English: Bengal gram split
   Hindi: Chane-ki-dal, Marathi: Harbara dal, Bengali: Cholar dal, Kashmiri: Chola dal. Tamil: Kadalai parappu, Malayalam: Kadala parappu, Telugu: Senaga pappu, Kannada: Kadale bele
3. English: Bengal gram roasted
   Hindi: Bhuna Chana, Marathi: Phutana, Bengali: Chola bhaja, Gujarati: Phutana, Tamil: Pottu kadalai, Malayalam: Varutha kadala, Telugu: Putnalu pappu, Kannada: Huri-kadale/puttani
4. English: Green gram whole
   Hindi: Mung, Marathi: Mung, Bengali: Mug, Gujarati: Mug, Punjabi: Moongi, Kashmiri: Muang, Oriya: Muga, Tamil: Pasipayir, Malayalam: Cheru payar, Telugu: Pesalu, Kannada: Hesaru kalu
5. English: Split and husked green gram
   Hindi: Mung dal/moong dal, Marathi: Mung dal, Bengali: Mug dal, Gujarati: Mug dal, Punjabi: Moongi, Kashmiri: Duli muang dal, Oriya: Muga dali, Tamil: Payaththam parappu, paasi paruppu Malayalam: Cheru payar parippu, Telugu: Pesara pappu, Kannada: Hesaru bele
6. English: Split yellow gram/pigeon pea
   Hindi: Arhar dal, Marathi: Tur dal/Toor dal, Bengali: Arhar dal, Gujarati: Tuver dal, Kashmiri: Arhar dal, Oriya: Arada, Tamil: Tuvaram paruppu, Malayalam: Tuvara parippu, Telugu: Kandi pappu, Kannada: Thugare bele
7. English: Split red gram
   Hindi: Masur dal, Marathi: Masur dal, Bengali: Masoor dal, Gujarati: Masur dal, Kashmiri: Musur, Oriya: Masura dali, Tamil: Masur paruppu, Malayalam: Masur parippu, Telugu: Misur pappu/ Erra kandi pappu, Kannada: Masur bele

Balanced diet
(Class 7)

Calorie value of select food items
Values given are for 100 gms of edible portions

Food	Calories Kcal	Proteins gms	Fats gms	Carbohydrates gms	Water gms	Vitamins
Milk – Cow	67	3.3	4.1	4.4	87	A, B2, Niacin
Milk – Buffalo	117	4.3	6.5	5	81	A, C, D & B12
Milk – Goat	72	3.3	4.5	4.6	86.8	A, D, E & B-1,2,3,6 & 12
Milk – Human	65	1.1	3.4	7.4	88	B12, D & K
Butter	740	–	82	–	15	A
Ghee – Cow	900	–	100	–	–	A, E
Cheese	310	22	25	–	72	A, B12
Ice Cream	170	4	7	25	64	A, B2, Niacin
Eggs	150	12	11	–	75	A, B1, B2, Niacin
Pork muscle	114	18.7	4.4	–	77.4	B2, Niacin
Fowl	109	25.9	0.6	–	77.2	
Fish (Hilsa)	273	21.8	19.4	2.9	53.7	B1, Niacin
Beef muscle (base on cuts)	114	22.6	5 12	–	74.3	B (complex) B2,3,6 and 12
Mutton muscle	194	18.5	13.3	–	71.5	B12, Niacin, B2 and 6
Shrimp	349	68.1	8.5	–	6	B12
Apple	59	0.2	0.5	13.4	84.6	C
Avocado pear	215	1.7	22.8	0.8	73.6	C, E, K & B-6
Banana ripe	116	1.2	0.3	27.2	70.1	C
Cherries	64	1.1	0.5	13.8	83.4	C
Dates dried	317	2.5	0.4	75.8	15.3	B complex, A, K
Grapes dark (blue variety)	58	0.6	0.4	13.1	82.2	C
Grapes green	71	0.5	0.3	16.5	79.2	C
Guava	51	0.9	0.3	11.2	81.7	D, K
Lychee	61	1.1	0.2	13.6	84.1	C
Mango ripe	74	0.6	0.4	16.9	81	A
Mulberry	49	1.1	0.4	10.3	86.5	C
Orange	48	0.7	0.2	10.9	87.6	C, A
Orange juice	9	0.2	0.1	1.9	97.7	Negligible content
Pappaya ripe	32	0.6	0.1	7.2	90.8	A
Peaches	50	1.2	0.3	10.5	86	C
Pears	52	0.6	0.2	11.9	86	C
Pineapple	46	0.4	0.1	10.8	87.8	C
Plum	52	0.7	0.5	11.1	86.9	A & K
Pomegranate	65	1.6	0.1	14.5	78	C, K, Folic acid
Sapota (Chiku)	98	1.1	0.5	21.4	73.7	A, C
Seethaphal (Custard apple)	104	1.6	0.4	23.5	70.5	C, B-6
Strawberry	44	0.7	0.2	9.8	87.8	C

Food	Calories Kcal	Proteins gms	Fats gms	Carbohydrates gms	Water gms	Vitamins
Cabbage	27	1.8	0.1	4.6	91.9	A, C
Coriander leaves	6.3	3.3	0.6	1.2	86.3	A
Spinach	26	2	0.7	2.9	92.1	A, B9 & C
Beetroot	43	1.7	0.1	8.8	87.7	Choline
Carrot	48	0.9	0.2	10.6	86	A
Colocasia (Arvi)	97	3	0.1	21.1	73.1	
Onion – Big	50	1.2	0.1	11.1	86.6	C
Onion small (shallots)	59	1.8	0.1	12.6	54.3	B6 & C
Radish Pink	32	0.6	0.3	6.8	90.8	C
Radiah White	17	0.7	0.1	3.4	94.4	C
Potato	97	1.6	0.1	22.6	74.7	C
Sweet Potato	120	1.2	0.3	28.2	68.5	C, B6
Yam	111	1.4	0.1	26	69.9	
Ash Gourd	10	0.4	0.1	1.9	96.5	C
Bitter Gourd	25	1.6	0.2	4.2	92.4	C, A, E, B1,2,3 & 9
Snake Gourd	18	0.5	0.3	3.3	94.6	A, B6, C & K in small amounts
Bottle Gourd	12	0.2	0.1	2.5	96.1	C, K
Brinjal	24	1.4	0.3	4	92.7	B, C & K
Broad beans	48	4.5	0.1	7.2	85.4	C & B
Peas green	93	7.2	0.1	15.9	72.9	C, K & B1
Cauliflower	30	2.6	0.4	4	90.8	C
Cluster beans	16	3.2	0.4	10.8	81	
Double beans	85	8.3	0.3	12.3	73.8	
Drumstick	26	8.3	0.1	3.7	86.9	C, A
Field Beans	48	3.8	0.7	6.7	86.1	
French Beans	26	1.7	0.1	4.5	91.4	
Ladies fingers	35	1.9	0.2	6.4	89.6	C, K
Pumpkin	25	1.4	0.1	4.6	92.6	A, C & E
Tomato – green	23	1.9	0.1	3.6	93.1	C, K
Green chillies	29	2.9	0.6	3	85.7	C, B6, K, A
Tamarind Pulp	283	3.1	0.1	67.4	20.9	
Ginger – fresh	67	2.3	0.9	12.3	80.9	
Garlic – dry	145	6.3	0.1	29.8	62	
Almond	655	20.8	58.9	10.5	5.2	
Cashewnut	596	21.2	46.9	22.3	5.9	
Coconut – Fresh	444	4.5	41.6	13	36.3	
Ground nut	567	25.3	40.1	26.1	3	Choline
Red gram	335	22.3	1.7	57.6	13.4	A, Niacin, B9, Choline
Rajmah	346	22.9	1.3	60.6	12	B9, B6, K & C
Bengal gram – whole	360	17.1	5.3	60.9	9.8	A, C, B complex and Choline

Food	Calories Kcal	Proteins gms	Fats gms	Carbohydrates gms	Water gms	Vitamins
Rice raw, milled	345	6.8	0.5	78.2	13.7	Niacin & folic acid
Rice parboiled, milled	346	6.4	0.4	79	13.3	Niacin & folic acid
Wheat flour	341	12.1	1.7	69.4	12.2	A, Niacin & Folic acid
Maida	348	11	0.9	73.9	13.3	A in samll amounts
Wheat bread	244	8.8	1.4	49	39	Negligible
White (maida) bread	245	7.8	0.7	51.9	39	Negligible
Ragi	328	7.3	1.3	72	13.1	A, Folic acid (B9)
Bajra	361	11.6	5	67.5	12.4	A, Niacin
Jowar	349	10.4	1.9	72.6	11.9	A, Niacin, Folic acid
Honey	319	0.3	0	79.5	20.6	Negligible
Jaggery	383	0.4	0.1	95	3.9	Negligible

NOTE: Blank cells indicate that the vitamin contents are in very small quantities.
One cup = 150 to 200 ml. Calorific value of a cup of tea = 30-90 Cal. Coffee = 20-100 Cal
1 Table spoon of sugar has a calorific value of 40-50 Cal

Parts of a microscope
(Class 8)

Teacher Resource Sheet

This sheet may be displayed on the class notice board, so that students can refer and learn to use the microscope correctly.

Students can check for correct answers allowing for self-study.

Download an image of a microscope with its parts labelled and put it up on the notice board.

How to get the microscope ready for viewing objects

1. First ensure that the objective lens and eyepiece are in line with one another. This is done by rotating the nosepiece gently so that the objective lens falls into the groove. A 'click' sound indicates that the objective lens is in place.
2. Next open the diaphragm fully.
3. If there is a lamp at the part marked light source then switch on the light source and ensure that when you see through the eyepiece there is bright light visible.
4. Some microscopes have a lens in place of the light source. In such cases adjust the lens such that natural light is reflected through the diaphragm onto the stage.
5. Lower the body tube completely. Make sure that you start working with 10 X objective lens and not the 45X or higher.

Now your microscope is set for viewing.

1. Keep the slide with the object on the stage and secure its edges with the clips.
2. Look through the eyepiece and slowly raise the body tube, using the coarse adjustment screw. When the object comes into view, stop moving the body tube. Use the fine adjustment knob to focus further such that you are able to see a sharp image.

For more on microscopes for the classroom see: https://biologyeducation.net/classrooms/the-hidden-world-of-microorganisms-1/

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